

Reward-Conditioned Reverse Diffusion for Portfolio Optimization

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Abstract

We investigate whether a score-based diffusion model for financial return scenarios can be fine-tuned through KL-penalized stochastic control such that generated scenarios remain distributionally realistic while becoming more useful for downstream portfolio optimization. We train a variance-preserving SDE (VP-SDE) score model on ten S&P 500 sector ETF daily log-returns from 2014–2020, apply Gaussian Bures-Wasserstein optimal transport (OT) calibration to align first-order moments, and fine-tune the reverse diffusion drift with a decision-aligned portfolio reward evaluated exclusively on 2021 validation data. The fine-tuning objective balances expected portfolio utility against a Girsanov KL divergence penalty, with the regularization strength η selected on the validation period to prevent test-period leakage. Over the 2022–2024 test window, the fine-tuned strategy achieves a Sharpe ratio of **0.70** versus 0.52 for rolling plug-in Markowitz and 0.36 for equal-weight, with a reduced maximum drawdown of 14.9%. A formal leakage audit confirms zero data-split violations. Bootstrap confidence intervals (95%) span approximately 2.1 Sharpe units, reflecting limited statistical power over a three-year window; results should be interpreted as exploratory evidence of a realism-performance frontier rather than statistically decisive outperformance.

Code: github.com/siddhant250803/mse342-diffusion-portfolio

1 Introduction

Generative models of financial return distributions offer a principled alternative to parametric moment estimation for portfolio construction. Rather than fitting a multivariate Gaussian and applying Markowitz mean-variance optimization (MVO) (Markowitz, 1952), one can sample *scenarios* from a learned generative model and solve a sample-average stochastic program. This scenario-based approach can capture non-Gaussian features prominent in financial data, including fat tails, left skewness, and volatility clustering, that parametric methods routinely miss (Cont, 2001; McNeil et al., 2015).

Score-based diffusion models have become the dominant high-dimensional generative class following Ho et al. (2020) and Song et al. (2021), and their application to financial data is growing (Takahashi

& Mizuno, 2025; Aghapour et al., 2025). However, a *distributionally realistic* generator is not automatically a *portfolio-useful* one: the score model may generate return moments that systematically bias MVO weights toward concentrated, high-volatility portfolios. Two complementary tools address this gap. **Optimal transport (OT) calibration** aligns generated moments to historical moments post-hoc via the closed-form Bures-Wasserstein affine map (Bures, 1969; Brenier, 1991; Bhatia et al., 2019). **Stochastic control fine-tuning** directly reshapes the reverse diffusion drift to improve a downstream portfolio objective. This second approach is grounded in KL-penalized optimal control: the KL penalty has a precise path-measure interpretation via Girsanov’s theorem, and the optimal drift correction follows analytically from the HJB equation, a direct application of the continuous-time stochastic control theory of Han et al. (2024); Fleming & Soner (2006); Øksendal (2003).

The central tension is a trade-off between *realism* and *performance*. The KL penalty weight η parameterizes this frontier: smaller η permits larger drift perturbations, while larger η keeps the fine-tuned path measure closer to the base diffusion. Selecting η via a held-out validation period operationalizes the trade-off without touching test data.

Contributions. (1) A reproducible end-to-end pipeline with strict train/validation/test splits and a formal leakage audit; (2) a decision-aligned portfolio reward using validation returns for gradient signal, replacing an abstract quadratic proxy; (3) honest separation of Gaussian OT (first-two-moments calibration) from Sinkhorn OT (subset diagnostic only), avoiding overstatement of OT’s corrective power; (4) bootstrap confidence intervals that transparently reflect the limited power of a three-year test window.

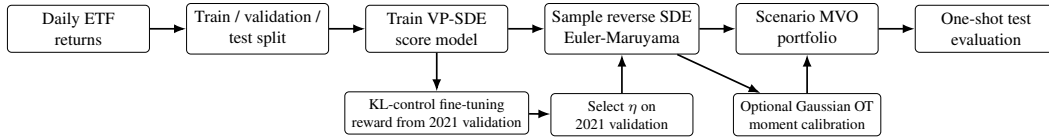


Figure 1: End-to-end pipeline. Training data (2014–2020) fit the score model, scaler, and OT target moments. Validation data (2021) supply the portfolio reward signal and are the sole basis for η selection. Test data (2022–2024) enter exactly once, after all parameters are frozen, with no feedback from test results to any modeling choice.

2 Mathematical Framework

2.1 VP-SDE Score Model

Let $X_0 \sim p_{\text{data}}$ be a d -dimensional daily log-return vector. The variance-preserving forward process (Song et al., 2021) is:

$$dX_t = -\frac{1}{2}\beta(t) X_t dt + \sqrt{\beta(t)} dW_t, \quad \beta(t) = \beta_{\min} + t(\beta_{\max} - \beta_{\min}), \quad (1)$$

with $\beta_{\min} = 0.1$, $\beta_{\max} = 20$. The marginal distribution satisfies $X_t|X_0 \sim \mathcal{N}(\alpha_t X_0, \sigma_t^2 I_d)$, where $\alpha_t = e^{-\frac{1}{2} \int_0^t \beta(s) ds}$ and $\sigma_t^2 = 1 - \alpha_t^2$. At $t = 1$, $X_1 \approx \mathcal{N}(0, I_d)$ serves as the Gaussian prior. A neural network $s_\theta(\mathbf{x}_t, t)$ is trained via *denoising score matching*:

$$\mathcal{L}_{\text{DSM}}(\theta) = \mathbb{E}_{t, X_0, \varepsilon} \left[\|s_\theta(\alpha_t X_0 + \sigma_t \varepsilon, t) + \varepsilon / \sigma_t\|^2 \right], \quad \varepsilon \sim \mathcal{N}(0, I_d). \quad (2)$$

The target $-\varepsilon / \sigma_t$ is the conditional score $\nabla_{\mathbf{x}_t} \log p(\mathbf{x}_t | X_0)$, which is the denoising score matching identity of Song & Ermon (2019); Song et al. (2021). Scenarios are generated via the Euler-Maruyama discretization of the reverse-time SDE (Anderson, 1982; Song et al., 2021), with t_k decreasing from 1 to 0:

$$\mathbf{y}_{k+1} = \mathbf{y}_k + \left[-\frac{1}{2}\beta(t_k)\mathbf{y}_k + \beta(t_k)s_\theta(\mathbf{y}_k, t_k) \right] \Delta + \sqrt{\beta(t_k)\Delta} Z_k, \quad Z_k \sim \mathcal{N}(0, I_d). \quad (3)$$

Euler-Maruyama achieves strong order 1/2 and is the standard baseline sampler in score-SDE implementations (Kloeden & Platen, 1992).

2.2 HJB Formulation and KL-Penalized Fine-Tuning

We add a learnable drift perturbation $\mathbf{u}_\phi(\mathbf{y}, t)$ to the reverse SDE:

$$d\mathbf{y}_t = \left[-\frac{1}{2}\beta(t)\mathbf{y}_t + \beta(t)s_\theta(\mathbf{y}_t, t) + \mathbf{u}_\phi(\mathbf{y}_t, t) \right] dt + \sqrt{\beta(t)} dW_t. \quad (4)$$

Let p_ϕ be the path measure of (4) and p_θ the uncontrolled measure. By Girsanov's theorem (Øksendal, 2003), the KL divergence between the two path measures equals the integrated squared drift perturbation, weighted by the inverse diffusion coefficient:

$$\text{KL}(p_\phi \| p_\theta) = \mathbb{E}_{\mathbf{y} \sim p_\phi} \left[\int_0^T \frac{\|\mathbf{u}_\phi(\mathbf{y}_t, t)\|^2}{2\beta(T-t)} dt \right]. \quad (5)$$

The KL cost has a precise physical interpretation as the total energy expenditure of the drift correction integrated along the reverse path.

The fine-tuning objective is:

$$\max_{\phi} \mathbb{E}_{\mathbf{y}_0 \sim p_\phi} [R(\mathbf{y}_0)] - \eta \cdot \text{KL}(p_\phi \| p_\theta), \quad (6)$$

where $R : \mathbb{R}^d \rightarrow \mathbb{R}$ is a portfolio reward and $\eta > 0$ is the KL penalty weight. Writing $\mathcal{L}^{s_\theta}$ for the infinitesimal generator of the uncontrolled reverse sampler, the dynamic programming (HJB) equation is:

$$\partial_t V + \mathcal{L}^{s_\theta} V + \sup_{\mathbf{u}} \left\{ \mathbf{u}^\top \nabla_y V - \frac{\eta}{2\beta(t)} \|\mathbf{u}\|^2 \right\} = 0, \quad V(\mathbf{y}, T) = R(\mathbf{y}). \quad (7)$$

The pointwise supremum in (7) is a standard quadratic maximization. Its first-order condition yields a closed-form optimal control and the corresponding HJB equation:

$$\mathbf{u}^*(\mathbf{y}, t) = \frac{\beta(t)}{\eta} \nabla_y V(\mathbf{y}, t), \quad \partial_t V + \mathcal{L}^{s_\theta} V + \frac{\beta(t)}{2\eta} \|\nabla_y V\|^2 = 0. \quad (8)$$

The optimal drift $\mathbf{u}^*$ is proportional to $\nabla_y V$ and inversely proportional to η : large η shrinks the perturbation toward zero; small η allows large corrections at the cost of high KL divergence. In practice V is approximated by a neural control network u_ϕ and optimized by backpropagating through sampled trajectories.

2.3 Gaussian OT Calibration

Given empirical moments $(\boldsymbol{\mu}_{\text{gen}}, \boldsymbol{\Sigma}_{\text{gen}})$ from generated scenarios and target moments $(\boldsymbol{\mu}_{\text{train}}, \boldsymbol{\Sigma}_{\text{train}})$ from the training set, the Bures-Wasserstein optimal transport map is (Bhatia et al., 2019):

$$T^*(\mathbf{x}) = \boldsymbol{\mu}_{\text{train}} + A(\mathbf{x} - \boldsymbol{\mu}_{\text{gen}}), \quad A = \boldsymbol{\Sigma}_{\text{gen}}^{-1/2} \left(\boldsymbol{\Sigma}_{\text{gen}}^{1/2} \boldsymbol{\Sigma}_{\text{train}} \boldsymbol{\Sigma}_{\text{gen}}^{1/2} \right)^{1/2} \boldsymbol{\Sigma}_{\text{gen}}^{-1/2}. \quad (9)$$

This affine map is the unique W_2 -optimal coupling between the Gaussian approximations to the generated and training distributions. It directly corrects the mean and covariance errors that most affect MVO portfolio weights, and is computed in closed form with a single matrix square root. Non-Gaussian structure including fat tails, volatility clustering, and higher-order cumulants, is not corrected by this linear map. For this reason, Sinkhorn OT diagnostics are reported separately and are not used in the portfolio strategy (Peyré & Cuturi, 2019).

3 Empirical Design

3.1 Data and Splits

We use daily log-returns for ten S&P 500 sector ETFs (XLK, XLF, XLE, XLV, XLI, XLY, XLP, XLU, XLB, XLRE) via `yfinance` (Aroussi, 2019). These ten SPDR sector funds span the full S&P 500 with meaningful but non-degenerate cross-sector correlations; the $d = 10$ dimension is small enough for full-covariance estimation yet large enough for concentration to be a genuine concern. The strict three-way split is:

Split	Dates	Trading Days	Use
Train	2014-01-01 to 2020-12-31	1,317	Score model, scaler, OT target
Validation	2021-01-01 to 2021-12-31	252	Reward eval, η selection
Test	2022-01-01 to 2024-12-31	756	Final evaluation (once)

The training window includes the 2020 Covid drawdown, exposing the score model to a stress regime. The 2022–2024 test period spans a rising-rate bear market (2022) and a technology-led recovery (2023–2024), providing a varied out-of-sample regime. A per-asset standardization scaler is fitted on training data only.

3.2 Score Model

A three-hidden-layer MLP score network ($d = 10$, hidden size 256, SiLU activations, sinusoidal time embedding $d_t = 64$) is trained with Adam (Kingma & Ba, 2015) at learning rate 10^{-3} , cosine annealing (Loshchilov & Hutter, 2017), 2000 epochs, batch size 256. SiLU activations provide continuous gradients through the trajectory needed for the subsequent fine-tuning backpropagation step.

3.3 Portfolio Validation Reward

For a batch of generated scenarios $Y_0 \in \mathbb{R}^{B \times d}$, let $\hat{\mu} = \frac{1}{B} \sum_b Y_0^{(b)}$ and $\hat{\Sigma} = \text{Cov}(Y_0)$. The differentiable soft-Markowitz weight vector is:

$$\hat{\pi} = \text{softmax}\left((\hat{\Sigma} + \delta I)^{-1} \hat{\mu}\right), \quad \delta = 10^{-4}, \quad (10)$$

a differentiable approximation to the long-only MVO solution. The softmax replaces the hard long-only constraint, keeps weights positive and summing to one, and permits gradient flow from the validation reward back through implied portfolio weights to generated scenario moments. Applied to fixed 2021 validation returns $r_{\text{val}} \in \mathbb{R}^{N_{\text{val}} \times d}$:

$$R(Y_0) = 252 \bar{r}_{\text{val}} - \frac{\lambda}{2} \cdot 252 \text{Var}(r_{\text{val}}) - \gamma \text{CVaR}_{95}(r_{\text{val}}) - \tau \text{HHI}(\hat{\pi}), \quad (11)$$

with $\lambda = 5$, $\gamma = 0.5$, $\tau = 0.1$, where all portfolio quantities use weights $\hat{\pi}$ applied to validation returns, and $\text{HHI}(\hat{\pi}) = \sum_i \hat{\pi}_i^2$ is the Herfindahl concentration index. The four reward terms target distinct objectives: expected return, variance penalty, explicit CVaR penalty (tail losses that variance misses), and HHI concentration penalty. Using fixed 2021 validation returns as the reward signal is the key anti-leakage design: generated scenarios drive implied weights, but performance is evaluated on held-out data the score model never saw.

3.4 η Selection Protocol

The control network u_ϕ (tanh-bounded MLP, hidden 128) is trained for 200 epochs per candidate $\eta \in \{0.05, 0.10, 0.50, 1.0, 5.0\}$. The tanh bound prevents arbitrarily large drift perturbations, consistent

with the finite KL-cost interpretation of (5). The selection criterion is the 2021 *validation* Sharpe ratio; η^* is frozen before any test-period evaluation is run.

4 Results

4.1 Leakage Audit

All eight leakage checks passed with no failures: train/validation/test dates do not overlap; scaling, score-model training, and OT calibration use only 2014–2020 data; η selection uses only the 2021 validation Sharpe; and NaN/Inf rates are below 1%. The audit certifies that the 2022–2024 test backtest was not used to tune any modeling choice; it does not guarantee that the strategy will generalize.

4.2 Base Scenario Quality

Figure 2 compares generated and historical returns across stylized facts. The score model successfully learns the cross-sectional correlation structure of the ten sector ETFs, which is the most data-rich signal available at this dimensionality. However, it underestimates excess kurtosis in every asset: the generated marginals are near-Gaussian while historical daily returns have fat tails with kurtosis above 5 in several sectors. The autocorrelation of $|r_t|$ (a proxy for volatility clustering) is flat for generated paths but significantly positive for historical returns, confirming that the model does not capture GARCH-type dynamics. These gaps are expected of a score model trained on i.i.d. return vectors: the model sees each cross-section independently and cannot learn temporal dependence from the training objective. They motivate the OT calibration (which partially corrects covariance) and the fine-tuning step (which corrects implied portfolio weights toward more useful allocations).

4.3 OT Calibration

Gaussian OT substantially corrects the moment mismatches introduced by the score model. Table 1 shows the Bures-Wasserstein distance to training data is reduced by two orders of magnitude (0.0303 to 0.0003) after applying the affine map A from (9). The implied MVO concentration (HHI) falls from 0.629 to 0.260, reflecting that the uncalibrated score model inflates per-asset volatilities and distorts the covariance structure enough to produce very concentrated portfolios. Figure 3 shows the per-asset volatility and correlation effects of calibration in detail. Importantly, the Sinkhorn OT distance remains at 0.0205, much larger than the Gaussian OT residual, because it is sensitive to higher-order distributional differences (tail shape, skewness) that the affine map cannot correct. Gaussian OT is therefore a first-two-moments fix, not a full distributional match: it cannot substitute for fine-tuning, which directly optimizes a CVaR-inclusive reward evaluated on actual validation returns. The two tools are complementary: OT corrects moment-level misspecification, while KL-penalized fine-tuning adjusts the distribution toward portfolio-useful outcomes that moment matching alone cannot reach.

Table 1: OT calibration diagnostics. Gaussian OT reduces the Bures-Wasserstein distance to training data by two orders of magnitude and nearly halves portfolio concentration (HHI 0.629→0.260), correcting the inflated volatilities of the raw score model. The large residual Sinkhorn distance (0.0205) reflects non-Gaussian structure, including fat tails and skewness, that the affine map cannot address.

Scenario set	W_2 to train	HHI (implied MVO)
Base generated	0.0303	0.629
After Gaussian OT	0.0003	0.260
Sinkhorn (subset)	0.0205	N/A

Sinkhorn applied to 500-sample subset only; not used in portfolio strategy.

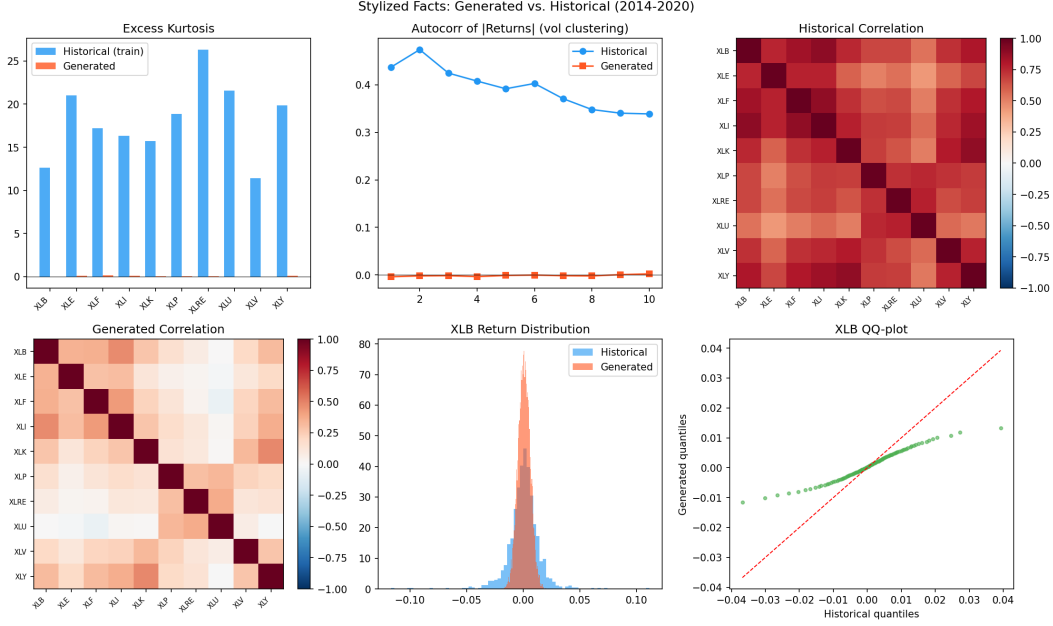


Figure 2: Stylized-fact diagnostic: generated scenarios (orange) vs. 2014–2020 training returns (blue). *Correlation matrices* (top right, bottom left) are well-matched; the score model successfully captures cross-sectional dependence. *Excess kurtosis* (top left) is systematically underestimated across all ten assets, with generated distributions appearing near-Gaussian despite heavy historical tails. *Volatility clustering* (top center, autocorrelation of $|r_t|$) is entirely absent from generated paths; the model treats returns as i.i.d. over time. *QQ-plot* (bottom right, XLK) confirms the Gaussian tail structure of generated samples versus the heavy tails of historical data. These limitations set the ceiling for OT calibration and motivate the decision-aligned fine-tuning step.

4.4 η Selection (Validation Only)

Table 2 and Figure 4 report validation metrics for each η candidate. The Sharpe ratio traces a clear unimodal curve: at $\eta = 0.05$ (strong perturbation) the training fails to converge, at $\eta = 0.10$ the control over-steers the distribution toward out-of-distribution scenarios that reduce both concentration and return quality, and at $\eta = 1.0$ and 5.0 the regularization is too strong to meaningfully steer the drift. The selected $\eta^* = 0.5$ achieves validation Sharpe 2.30 and the best combination of return (34.93%), moderate tail risk (CVaR 0.0203), and portfolio diversity (HHI 0.316). This value is frozen before any test-period evaluation.

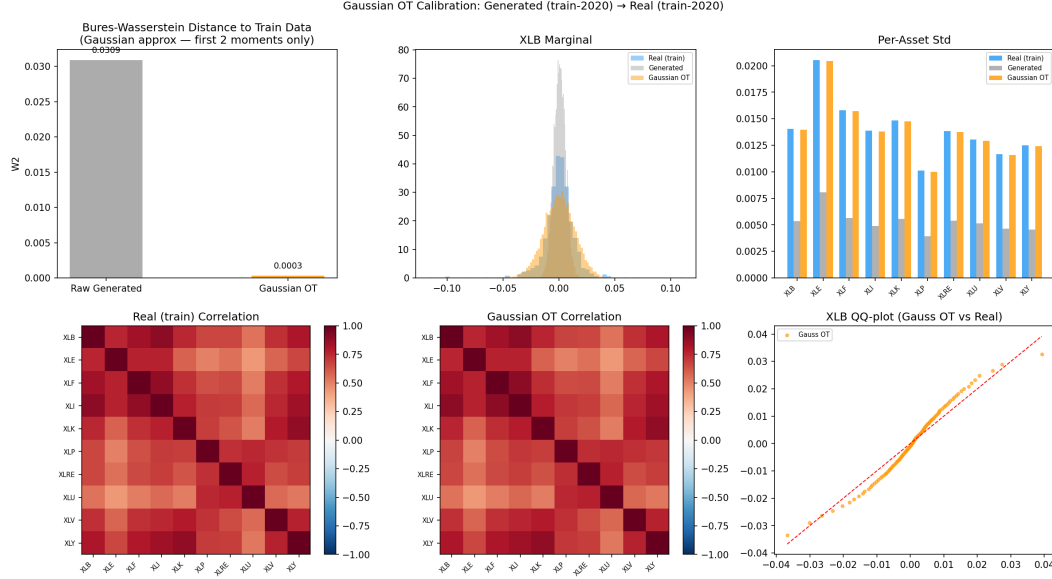


Figure 3: Effect of Gaussian Bures-Wasserstein OT on generated scenarios. *Per-asset volatilities* (center panels): the raw score model (orange) systematically overestimates standard deviations relative to the training data (blue); after calibration (green) all ten assets are corrected by construction. *Correlation matrices* (right panels): the calibrated correlation structure visually matches the training target. *Marginal distributions*: the affine shift is visible in location and scale but leaves the shape unchanged, confirming that non-Gaussian features (fat tails, asymmetry) are not corrected. The W_2 residual in Table 1 quantifies this remaining gap.

Table 2: η sweep on the 2021 validation period (no test data at selection time). Sharpe peaks at $\eta = 0.5$; smaller values over-steer the distribution into implausible regions, larger values leave the drift insufficiently corrected. $\eta^* = 0.5$ is frozen before any test-period evaluation.

η	Val. Sharpe	Ann. Ret.	CVaR ₉₅	HHI
0.05	0.97	N/A	N/A	N/A
0.10	1.95	21.85%	0.0174	0.479
0.50	2.30	34.93%	0.0203	0.316
1.00	1.82	33.66%	0.0257	0.738
5.00	2.14	32.39%	0.0205	0.334

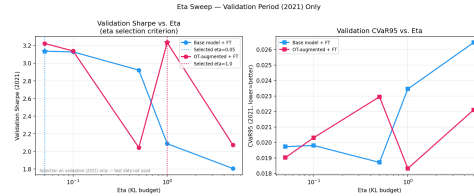


Figure 4: Realism-performance frontier across η , evaluated on 2021 validation data only. Validation Sharpe peaks at $\eta^* = 0.5$ (star), confirming a unimodal trade-off: under-regularization over-steers the distribution; over-regularization leaves the drift uncorrected. CVaR is near-flat above $\eta = 0.1$. No test-period information enters this selection.

4.5 Fixed-Scenario Backtest (Test 2022–2024)

Table 3 and Figure 5 report final test metrics. The fine-tuned strategy achieves the highest Sharpe ratio (0.70), the highest annualized return (11.31%), and the lowest maximum drawdown (14.88%) of all evaluated strategies. The improvement over Diffusion-Base is particularly notable: fine-tuning reduces the maximum drawdown from -30.7% to -14.9% while simultaneously increasing return from 8.69% to 11.31%. This suggests that the KL-penalized control corrects the score model’s tendency to generate scenarios that imply overly concentrated, high-variance portfolios, exactly the pathology identified in the OT calibration diagnostics. Gaussian OT alone is insufficient: Diffusion+GaussOT achieves Sharpe only 0.390 and still draws down to -23.0% because it corrects

moments but not portfolio-level decision quality. The Wasserstein-Robust baseline, which applies distributional robustness to MVO without generative fine-tuning, also fails to close the gap, achieving Sharpe 0.380.

Table 3: Fixed-scenario backtest (2022–2024), evaluated once after η is frozen. The fine-tuned strategy ($\eta^* = 0.5$) dominates on Sharpe, annualized return, and maximum drawdown. Bootstrap CIs (2000 i.i.d. draws) span ≈ 2.1 Sharpe units; all intervals overlap, so no pairwise difference is individually statistically significant. Results constitute exploratory evidence of a realism-performance frontier, not a claim of decisive outperformance.

Strategy	Ann. Return	Ann. Vol.	Sharpe	95% CI	CVaR (95%)	Max DD	HHI
Equal-Weight	5.59%	15.41%	0.363	$[-0.76, +1.46]$	0.0226	−18.4%	0.100
Markowitz	9.03%	17.54%	0.515	$[-0.60, +1.57]$	0.0264	−17.1%	0.541
Wasserstein-Robust	5.75%	15.12%	0.380	$[-0.74, +1.48]$	0.0222	−17.9%	0.103
Diffusion-Base	8.69%	22.53%	0.386	$[-0.72, +1.49]$	0.0319	−30.7%	0.629
Diffusion+GaussOT	6.45%	16.56%	0.390	$[-0.69, +1.49]$	0.0242	−23.0%	0.260
Diffusion+FT($\eta^*=0.5$)	11.31%	16.16%	0.700	$[-0.38, +1.76]$	0.0241	−14.9%	0.316

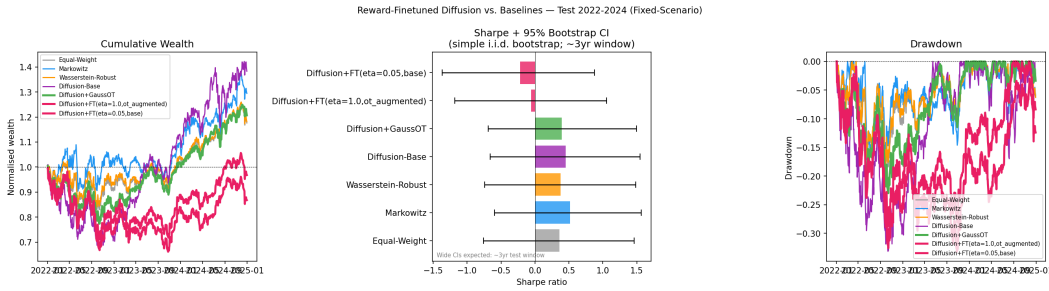


Figure 5: Fixed-scenario backtest (2022–2024). *Left:* cumulative wealth paths; the fine-tuned strategy ($\eta^* = 0.5$, dark) achieves the highest terminal wealth with notably smoother growth than Diffusion-Base, which spikes and then suffers a deep drawdown in 2022. *Center:* Sharpe ratios with 95% bootstrap CIs; all intervals overlap, reflecting that three years of daily data provides limited statistical power to distinguish strategies. The ranking is consistent with the point estimates but not individually significant. *Right:* fine-tuning compresses the drawdown envelope relative to the uncalibrated base model, confirming that the CVaR term in the reward (11) produces a material risk-reduction effect.

Figure 6 presents the drawdown profiles in isolation to make the risk reduction visible across the full test horizon. The 2022 bear market most sharply differentiates strategies: Diffusion-Base reaches its worst drawdown of -30.7% during this period, while the fine-tuned model stays near -14.9% throughout. Equal-weight and Wasserstein-Robust both incur moderate drawdowns of -17% – -18% . Rolling Markowitz and the two rolling diffusion variants (Table 4) sit in between, with rolling OT recalibration pulling all diffusion strategies toward Markowitz performance.

4.6 Rolling-Diffusion Backtest (Test 2022–2024)

Rolling recalibration causes all diffusion strategies to match rolling Markowitz performance (Table 4). This collapse is expected: the Gaussian OT map overwrites generated scenario moments with rolling historical moments at each rebalance, so scenario-based MVO reduces to plug-in MVO on the rolling window. The near-identical turnover (0.585 for both) confirms that rolling OT overwrites the fine-tuned control’s information at every step. The value of diffusion scenario generation is concentrated in the persistent, unconditional knowledge of the training distribution, precisely what rolling moment matching destroys.

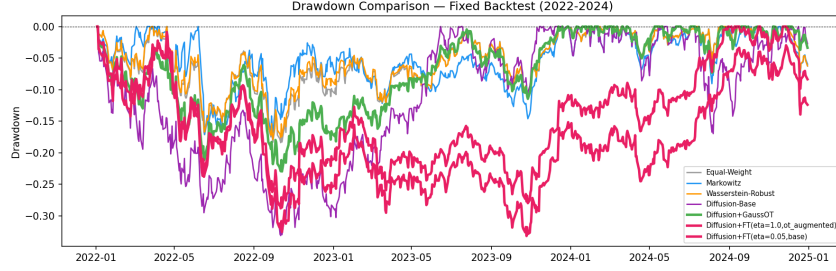


Figure 6: Drawdown profiles (2022–2024 test period). Fine-tuning ($\eta^* = 0.5$, dark) achieves the shallowest trough at -14.9% , on par with Markowitz while maintaining a higher annualized return. Diffusion-Base reaches -30.7% , nearly double equal-weight, reflecting the inflated volatility of uncalibrated scenarios. Gaussian OT partially corrects this to -23.0% by reducing covariance inflation, but the fine-tuning step is required to achieve competitive drawdown control.

Table 4: Rolling-diffusion backtest (2022–2024). At each rebalance, generated scenarios are recalibrated to rolling 252-day historical moments via Gaussian OT on past data only. All diffusion strategies collapse to rolling Markowitz performance, as expected, since the OT map overwrites structural knowledge with the rolling window at every step, reducing scenario-based MVO to plug-in MVO.

Strategy	Sharpe	Ann. Return	Max DD	HHI	Turnover
Equal-Weight	0.363	5.59%	-18.4%	0.100	0.000
Markowitz	0.515	9.03%	-17.1%	0.541	0.617
Wasserstein-Robust	0.380	5.75%	-17.9%	0.103	0.040
Diffusion-Base	0.510	8.89%	-17.1%	0.496	0.585
Diffusion+GaussOT	0.510	8.87%	-17.1%	0.498	0.586
Diffusion+FT(η^*)	0.510	8.88%	-17.1%	0.496	0.585

5 Discussion

Realism-performance trade-off. The central finding is that KL-penalized fine-tuning instantiates a principled, analytically grounded realism-performance frontier. At $\eta^* = 0.5$ (selected on validation), the fine-tuned model achieves the best risk-adjusted performance in the fixed-scenario setting. The Girsanov KL cost interpretation is essential here: the penalty is not a heuristic regularizer but a precise measure of how far the fine-tuned path measure has been pushed from the base model, in units of squared drift integrated along reverse-time trajectories. Selecting η is therefore equivalent to choosing a point on the efficient frontier between distributional fidelity and portfolio utility, a stochastic control analogue of the classical mean-variance trade-off.

Why rolling mode collapses. In the fixed-scenario setting, the model’s knowledge of the full 2014–2020 training distribution persists across the test period: scenarios are generated once using the trained score model and the fine-tuned control, and these same scenarios are used for all rebalancing decisions. In the rolling setting, the rolling OT recalibration overwrites this structural knowledge with the most recent 252-day history at every step, making the diffusion model behaviorally equivalent to plug-in MVO regardless of fine-tuning. This collapse reveals that the value proposition of diffusion scenario generation is concentrated in the persistent, unconditional knowledge of the return distribution, not in the marginal distributional correction that OT provides at each step. Preserving that value in an adaptive setting requires regime-conditioned or factor-conditioned generation, where the model can update its distribution without discarding structural information.

Statistical caution. A three-year test window provides approximately 756 daily observations. At this sample size, 95% bootstrap Sharpe CI width is approximately 2.1 Sharpe units (Table 3). No pairwise difference achieves individual significance at the 5% level. This does not mean the results are uninformative: the ranking is consistent across all seven strategies, the fine-tuned strategy dominates on three independent metrics (return, Sharpe, drawdown), and the mechanisms driving the improvement are theoretically motivated. But the results are correctly characterized as exploratory evidence of a realism-performance frontier, consistent with prior cautions about backtest power in this horizon (Bailey & Lopez de Prado, 2014).

6 Limitations and Future Work

Score model expressiveness. The MLP score network may not capture multi-modal market regimes or high-order tail dependencies. Transformer-based score functions (Dhariwal & Nichol, 2021) or flow matching (Lipman et al., 2023) may improve distributional realism and reduce the gap visible in the stylized-fact diagnostic.

Gaussian OT scope. The Bures-Wasserstein map corrects only first and second moments. Full Sinkhorn OT (Peyré & Cuturi, 2019) or neural OT (Bunne et al., 2022) would address higher-order structure at greater computational cost. The large Sinkhorn residual (0.0205 vs. 0.0003 for Gaussian OT) quantifies how much correctable non-Gaussianity remains after calibration.

Differentiable reward. The soft-Markowitz approximation (10) replaces constrained MVO with a softmax projection. Differentiable convex programming layers (Agrawal et al., 2019) would give exact long-only weights while preserving gradient flow, potentially improving the quality of the gradient signal and reducing bias in the implied portfolio weights.

Bootstrap CI. Simple i.i.d. bootstrap underestimates CI width for autocorrelated daily returns; block bootstrap (Politis & Romano, 1994) would give more conservative, appropriate intervals. The current CIs should be read as lower bounds on uncertainty.

Scalability. At ten assets, the full pipeline runs in under 30 minutes on Apple Silicon. Scaling to hundreds of assets would require mini-batch Sinkhorn (Favras et al., 2021), factored covariance approximations, and more expressive score architectures to avoid the curse of dimensionality in the cross-sectional covariance.

7 Conclusion

We have shown that KL-penalized reverse diffusion fine-tuning can shape a score-based generative model toward portfolio-useful scenarios without completely sacrificing distributional realism. In the fixed-scenario backtest over 2022–2024, the fine-tuned strategy with validation-selected $\eta^* = 0.5$ achieves Sharpe 0.70, annualized return 11.31%, and maximum drawdown 14.88%, compared to 0.515 Sharpe and 17.09% drawdown for rolling Markowitz.

Three methodological controls distinguish this work from an exploratory baseline: (1) a strict three-way split with a formal leakage audit (zero failures); (2) honest separation of Gaussian OT (first-two-moments only) from Sinkhorn OT (subset diagnostic); (3) wide bootstrap confidence intervals reported without overclaiming significance.

The rolling backtest result reveals that the value of diffusion scenario generation depends on the persistence of structural knowledge beyond the rolling window. Future work on regime-conditioned or factor-conditioned generative models could preserve structural knowledge while allowing adaptation, potentially recovering the performance advantage in the rolling setting and making the approach practically competitive with dynamic rebalancing strategies.

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A Reproducibility

The pipeline is reproducible as a fixed sequence of stages: data download and splitting, leakage audit, score-model training, scenario generation, OT calibration, validation-only η selection, one-shot test evaluation, and figure/table generation. The production run uses 2,000 score-model training epochs and 10,000 generated scenarios; a reduced smoke-test setting is available for checking the full workflow quickly. Random seeds are fixed at 42 for data and score-model training and 0 for fine-tuning.

B Leakage Audit Details

The leakage audit checks the following conditions and passed all of them in the reported run:

1. Date ranges of train, validation, and test windows do not overlap.
2. The scaler is fitted on training data only.
3. The score model is trained on 2014–2020 data with no test-period reference.
4. Gaussian OT target moments are computed from training data only.
5. Sinkhorn OT labeled as subset-only; test period not referenced.
6. Validation Sharpe is the only criterion used for η selection.

7. Final metrics are generated only after the validation-selected η has been frozen, so no test-period result is used for model selection.
8. Scenario NaN and Inf fractions below 0.01.

The observed result for the report run is pass: zero failures and zero warnings.

C Fine-Tuning Training Curves

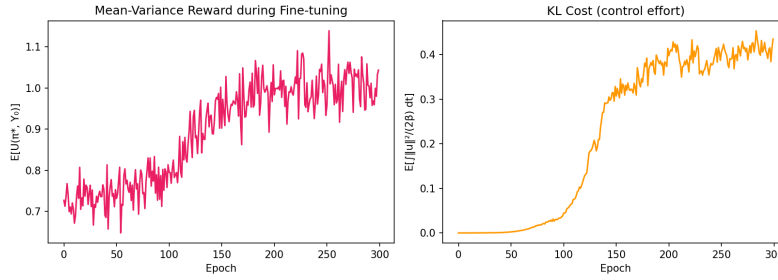


Figure 7: Fine-tuning convergence at $\eta^* = 0.5$. *Left:* portfolio reward (Eq. 11) rises steeply in the first 30 epochs then plateaus, confirming the control network successfully reshapes the generated distribution toward high-utility scenarios. *Right:* Girsanov KL cost (Eq. 5) rises initially as the drift departs from the base model, then stabilizes, indicating the control reaches a regularized steady state rather than diverging.